

# Healthcare Worker Burnout as a Determinant of Patient Care Quality: A Systematic Review

Mallarshini Naganthran<sup>1</sup>, Ashwini Naganthran<sup>2</sup>

<sup>1</sup> Emergency Department, Hospital Kajang, Selangor, Malaysia

<sup>2</sup> Universiti Putra Malaysia, Selangor, Malaysia

## INTRODUCTION



Burnout among healthcare workers is increasingly recognized as a major occupational health concern arising from chronic workplace stress.

It can lead to emotional exhaustion, reduced concentration, detachment, decreased empathy, and lower professional effectiveness, which may contribute to medical errors, compromised patient safety, and reduced patient satisfaction (Maslach and Jackson 1981; Saivagionl et al. 2017).

Although many studies have examined burnout and patient outcomes, evidence on how burnout influences overall quality of care and effective mitigation strategies remains fragmented.

This systematic review therefore examines the relationship between healthcare worker burnout and patient care quality and explores potential strategies to reduce its adverse effects on healthcare professionals and patients.

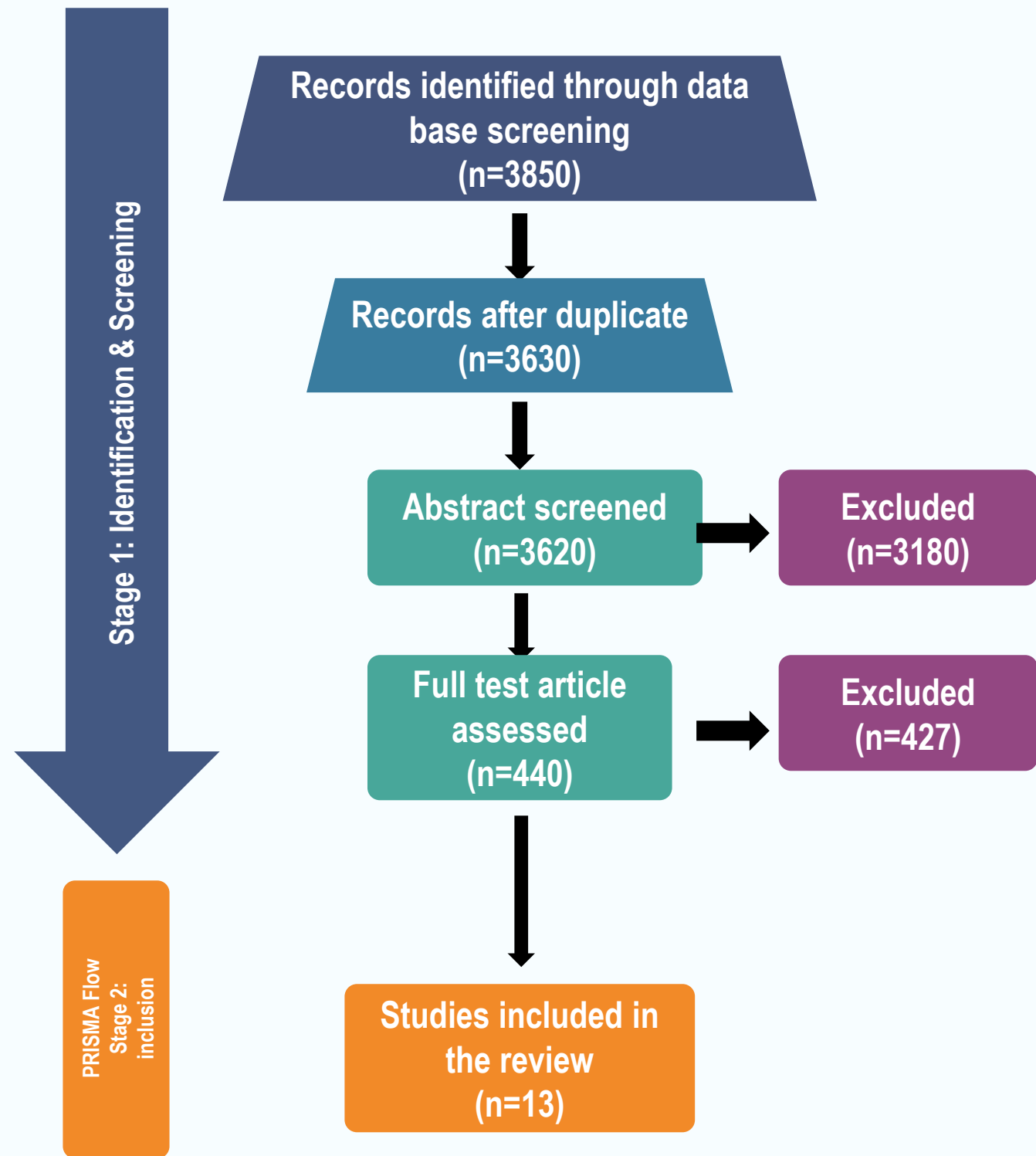
## METHODS



A systematic literature search was conducted for studies published between 2010 and 2023 using ISI Web of Knowledge, PubMed, Scopus, Ovid Medline, and Google Scholar, following PRISMA guidelines.

Two reviewers independently screened titles, abstracts, full tests, and reference lists. Studies using validated burnout or stress measures and examining patient care outcomes among healthcare professionals were included.

Studies without empirical correlation data, focusing only on non-organizational factors, or lacking separate healthcare professional analyses were excluded. Findings were interpreted using Conservation of Resources theory.



## RESULTS

440 records

The database search identified 440 records, of which 13 studies met the eligibility criteria.

Included studies were conducted across the Philippines, China, Italy, the United States, Ethiopia, and Portugal, with sample sizes ranging from 325 to 20,496 participants.



Higher burnout was associated with reduced patient satisfaction, impaired therapeutic communication, poorer perceived quality and safety of care, job dissatisfaction, and greater turnover intention.

Burnout prevalence reached 58% among emergency nurses, while physician burnout increased from 44.4% to 50.4% over four years.



Organizational determinants included heavy workload, suboptimal practice environments, and inadequate workplace support, whereas supportive work environments and stronger work engagement were associated with more favorable staff and patient outcomes.

### Summary of Included Studies

| Main Author | Sample | Main Findings                                                                                                                                                    | Resources / Threat-Loss / Conservation-Gain                                                        |
|-------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Folguera    | 549    | Lower work engagement was associated with increased burnout and turnover intention, while higher dedication was linked to better perceived patient care quality. | Resource: Work engagement<br>Threat/Loss: Job burnout<br>Gain: Quality care                        |
| Na          | 342    | Depression, low compassion satisfaction, burnout, and compassion fatigue were identified among emergency healthcare workers.                                     | Resource: Mental resilience<br>Threat/Loss: Depression<br>Gain: Stress management                  |
| Perego      | 325    | Stress, depression, anger, and exhaustion increased, while support networks and work conditions acted as protective factors.                                     | Resource: Mental resilience<br>Threat/Loss: Depression<br>Gain: Stress management                  |
| Muir        | 7,678  | Burnout was reported in 58%, job dissatisfaction in 38%, and intention to leave in 27%; better work environments improved care and safety.                       | Resource: Supportive environment<br>Threat/Loss: Turnover intention<br>Gain: Quality care          |
| Ortega      | 1,373  | Physician burnout increased from 44.4% to 50.4% over four years, with persistent burnout in 28.7% of participants.                                               | Resource: Supportive environment<br>Threat/Loss: Turnover intention<br>Gain: Quality care          |
| Plantinga   | 1,249  | One-third of staff met burnout criteria; heavy workloads and poor patient care environments were key contributing factors.                                       | Resource: Supportive environment<br>Threat/Loss: Turnover intention<br>Gain: Engagement strategies |
| Mersho      | 408    | Burnout significantly reduced effective therapeutic communication among nurses.                                                                                  | Resource: Supportive environment<br>Threat/Loss: Turnover intention<br>Gain: Engagement strategies |
| Ortega      | 1,373  | Nursing practice environment was associated with burnout, care quality, and patient safety.                                                                      | Resource: Supportive environment<br>Threat/Loss: Depression<br>Gain: Engagement strategies         |
| Lucas       | 1,859  | Burnout negatively influenced patient satisfaction, while better work conditions were linked to higher satisfaction.                                             | Resource: Supportive environment<br>Threat/Loss: Depression<br>Gain: Quality care                  |
| Vahey       | 828    | Burnout was common and was associated with clinician turnover and poorer workforce outcomes.                                                                     | Resource: Supportive environment<br>Threat/Loss: Depression<br>Gain: Quality care                  |
| Willard     | 748    | Burnout was significantly associated with lower patient satisfaction; better work environments reduced burnout.                                                  | Resource: Work engagement<br>Threat/Loss: Turnover intention<br>Gain: Engagement strategies        |
| Brooks      | 463    | Improved work environments reduced the negative effects of nurse burnout on patient outcomes.                                                                    | Resource: Supportive environment<br>Threat/Loss: Depression<br>Gain: Quality care                  |
| Amelia      | 20,496 | Greater work engagement and supportive environments were associated with lower burnout, reduced turnover, and improved patient outcomes.                         | Resource: Supportive environment<br>Threat/Loss: Depression<br>Gain: Stress management             |

## DISCUSSION



The findings are consistent with the Conservation of Resources (COR) theory, which proposes that burnout develops when healthcare workers experience actual or threatened loss of valued personal and professional resources (Hobfoll 1989).

Work engagement, mental resilience, and supportive work environments act as protective resources that reduce burnout and sustain patient care quality (Schaufeli and Bakker 2004; Mealer et al. 2012).

Conversely, depression and turnover intention reflect resource depletion and may contribute to reduced empathy, impaired communication, medical errors, and poorer patient outcomes (Bianchi et al. 2015; Fahrenkopf et al. 2008).



Resource conservation and gain can be promoted through supportive leadership, flexible work schedules, mindfulness, resilience training, professional autonomy, recognition, and staff engagement strategies (Shapiro et al. 2005; Garcia-Sierra et al. 2016).

Strengthening these resources may improve job satisfaction, reduce turnover, enhance patient safety, and support sustained delivery of high-quality care (Aiken et al. 2002; Green et al. 2020).

## CONCLUSION

Burnout mitigation requires integrated individual and organizational strategies that strengthen work engagement, resilience, staffing adequacy, professional development, and supportive workplace cultures. Collaborative implementation of these measures, alongside scalable digital stress-management interventions, may enhance healthcare worker well-being and improve patient care quality.

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